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Reducing emissions from deforestation and forest degradation (REDD+): game changer or just another quick fix?

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Reducing emissions from deforestation and forest degradation (REDD+) provides financial compensation to land owners who avoid converting standing forests to other land uses. In this paper, we review the main opportunities and challenges for REDD+ implementation, including expectations for REDD+ to deliver on multiple environmental and societal cobenefits. We also highlight a recent case study, the Norway–Indonesia REDD+ agreement and discuss how it might be a harbinger of outcomes in other forest-rich nations seeking REDD+ funds. Looking forward, we critically examine the fundamental assumptions of REDD+ as a solution for the atmospheric buildup of greenhouse gas emissions and tropical deforestation. We conclude that REDD+ is currently the most promising mechanism driving the conservation of tropical forests. Yet, to emerge as a true game changer, REDD+ must still demonstrate that it can access low transaction cost and high-volume carbon markets or funds, while also providing or complimenting a suite of nonmonetary incentives to encourage a developing nation's transition from forest losing to forest gaining, and align with, not undermine, a globally cohesive attempt to mitigate anthropogenic climate change.

Keywords: carbon payment scheme; biodiversity conservation; climate change; REDD; tropical deforestation

Introduction

The growing world population is driving global demand for food, feed, fuel, and other raw materials. These demands create economic opportunities—mediated by commodity markets and trade policies—that many developing countries respond to by exploiting and exporting their natural resources. As a consequence, vast areas of tropical forests are being cleared for agriculture, timber extraction, oil and gas development, and mining and infrastructure expansion.

Recognizing the economic impetus behind these exploitative activities, conservationists have been developing and implementing "Payments for Ecosystem Service" (PES) schemes, which provide financial compensation to land owners who avoid converting standing forests to other land uses.[1] These schemes typically involve monetization of key environmental services that society values and that forests might provide. The most prominent of such PES schemes is Reducing Emissions from Deforestation and Forest Degradation plus the conservation, sustainable management, and enhancement of forest carbon stocks (REDD+).

What is REDD+?

REDD+ is simply a method for putting a price tag on the carbon storage and sequestration services provided by forests. The basic premise of REDD+ is that, without it, a certain amount of carbon dioxide would be emitted due to the loss or degradation of forests. These expected emissions serve as a reference level to measure efforts to protect forests and reduce carbon emissions. If realized emissions are below the reference level, these reductions are considered "additional," and a corresponding amount

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of carbon credits may be awarded. To be sold on carbon markets, carbon credits must be generated through a rigorous process of measuring, reporting, and verification (or MRV), which might be subjected to certification by recognized standards, such as the Verified Carbon Standard (www.v-c-s.org). Early REDD+ demonstration and early action projects have been largely site based, protecting a particular tract of forests,[2] whereas for full-scale REDD+ implementation, the MRV process and financial compensation will take place at the national scale.[3]

The "+" after REDD comes from more recent discussions that have broadened the mechanism's scope to also recognize the carbon benefits of forest conservation, improved forest management, and the sequestration potential of afforestation and reforestation. At the same time, expectations for REDD+ to deliver on multiple environmental and societal issues have risen. These "cobenefits" might include conservation of biodiversity and improvements to rural livelihoods.

Much of the hype surrounding REDD+ stems from the perceived scale of financing,[4] yet this financing remains surprisingly nebulous. Almost all the demand for carbon credits comes from what are known as compliance carbon markets,[5] which are markets that sell credits for meeting obligations under the United Nations Framework Convention on Climate Change (UNFCCC). The UNFCCC is an international treaty produced at the United Nations Conference on Environment and Development (or Earth Summit) in Rio de Janeiro in 1992. The Kyoto Protocol to the UNFCCC has been ratified by 37 industrialized countries who commit themselves to a reduction of greenhouse gas (GHG) emissions by 5.2% from 1990 baseline levels. To help these "Annex I" countries meet their commitments, they are allowed to offset emissions by purchasing certified emission reduction (CER) credits, each equivalent to the reduction of one metric ton of carbon dioxide. These CER credits can be generated from the implementation of Clean Development Mechanism (CDM) projects in developing countries, which lead to net emission reductions. These CDM projects typically involve production of renewable energy (e.g., hydro-, thermal, or wind power) and improvements in the handling of industrial and agricultural byproducts.

When REDD+ (at the time it was just RED) was first proposed in 2005 by the Coalition for Rainforest Nations (www.rainforestcoalition.org) as a strategy for correcting the market failures that lead to rampant deforestation, it was envisaged that demand for REDD+ credits would be created, and conservation funds generated, from REDD+'s inclusion as a legitimate emissions reduction activity of the CDM. Despite intense lobbying by proponents of REDD+—particularly by the World Bank's Forest Carbon Partnership Facility (http://wbcarbonfinance.org/) and the United Nations–REDD Program (http://un-redd.org)—REDD+ is still not recognized as a legitimate CDM activity. As REDD+ negotiations are now taking place as part of long-term cooperative action under the UNFCCC, rather than the Kyoto Protocol, REDD+ is unlikely to ever be incorporated in the CDM.

Despite this limitation, REDD+ initiatives are being financed and implemented across the developing world. In their review of 64 developing countries, Cerbu *et al.*[6] found that as of October 2009, there were at least 79 REDD+ readiness activities (such as monitoring carbon stocks and identifying sources and drivers of emissions) and 100 REDD+ demonstration activities, largely in Indonesia (21) and Brazil (17). Some of the finance for these early initiatives comes from voluntary carbon markets, such as the Chicago Climate Exchange (www.chicagoclimatex.com), where credits are bought by companies that voluntarily wish to offset their emissions. Most of the funding, however, comes through voluntary carbon funds, many of them linked to the REDD+ Partnership, which has already helped to channel almost US $8 billion for REDD+, and bilateral agreements, such as the Norway–Indonesia REDD+ deal involving $1 billion.

It appears, for now at least, that REDD+ has gained momentum and access to financing independent of the UNFCCC process. The REDD+ partnership (http://reddpluspartnership.org), which includes 72 partner countries, is acting as the interim platform for REDD+ financing, policy development, and knowledge sharing. Yet, the continued growth and success of REDD+ still hangs in the balance. It depends not only on the mechanisms eventual formal inclusion in a post-Kyoto climate agreement, but also on its perceived ability to deliver meaningful climate benefits and surmount ongoing challenges.

In this paper, we review the main opportunities and challenges for REDD+ implementation. We also highlight a recent case study: the Norway–Indonesia REDD+ agreement. Looking forward, we critically examine the fundamental assumptions of REDD+ as a solution for the atmospheric buildup of GHG emissions and tropical deforestation and discuss recent proposals to expand the scope of REDD+ in terms of both the land uses it encompasses and its revenue sources.

Promises and opportunities

Emissions reductions

By reducing negative changes in forest carbon stores and promoting positive changes through forest restoration, reforestation, and afforestation, the first and most important promise of REDD+ lies in its potential to mitigate anthropogenic climate change. While subtropical and temperate forests in developing countries are technically eligible, tropical forests have emerged as the central candidate for REDD+ implementation due to their high carbon content and rates of forest loss, as well as their heightened potential for carbon sequestration.[7,8]

In 2005, tropical humid forests covered roughly 11 million km^2 and tropical dry forests covered a further 7.5 million km^2[9] combined, storing 248 Gt of biomass carbon in the vegetation,[10] which is roughly equivalent to the total fossil fuel emissions since the industrial revolution.[11] When these forests are cleared or logged, much of this carbon is quickly released into the atmosphere through fires and decomposition.[12]

When REDD+ was first proposed, it was thought that tropical deforestation was responsible for 17–25% of anthropogenic carbon emissions.[11,13,14] Since then, these figures have been revised downward, partly because rates of deforestation are lower than previously estimated,[15–18] and partly because fossil fuel emissions have increased.[19] The most recent estimates indicate that through the early 2000s, net deforestation and forest degradation released 1.22–1.30 PgC per year,[7,20] plus a further 0.30 PgC per year from peat degradation,[20] which is closely associated with deforestation.[21] This makes tropical deforestation and forest degradation responsible for roughly 15% of all anthropogenic carbon emissions, and in turn, this is the theoretical maximum by which REDD+ could reduce carbon emissions.

Global carbon markets traded a staggering $142 billion in 2010.[5] If REDD+ is connected to compliance markets, a portion of these funds is potentially available to help finance the protection of forests and the reduction in forest-based emissions. Even if REDD+ is instead financed through carbon funds, the size of early funds, such as those associated with the REDD+ partnership, indicates that REDD+ will have access to considerable financing to tackle tropical deforestation.

To look at it another way, forests could play a major role in reducing the costs and increasing the effectiveness of climate mitigation efforts. The Eliasch review,[22] the follow-up to the Stern Review,[23] estimates that including REDD+ in carbon markets could provide financing to reduce forest-based emissions by 75% by 2030; taken together with afforestation and reforestation, this would make forests carbon neutral. Because REDD+ appears to be a relatively cost-effective mechanism for reducing emissions, especially when compared with options such as carbon capture and storage,[23] including forests could lower the overall costs of meeting global emissions reductions targets. If REDD+ is included in a post-Kyoto climate deal, the cost of reducing global carbon emissions to half of 1990 levels would be reduced by 50% in 2030 and 40% in 2050.[22] These reduced costs could be taken as a means to make climate mitigation more palatable to the pocket-wise voter or instead as an opportunity to pursue more ambitious targets at no additional cost.

Forest and biodiversity conservation

At least half of all known and yet undiscovered species are found in tropical forests,[24–26] and their widespread loss is one of the major drivers of the current loss of global biological diversity.[27–31] Consequently, 18 of 25 global priorities for the conservation of imperiled biodiversity, or "biodiversity hotspots," are tropical forest regions.[32] Despite the importance of these areas for conservation, there has arguably been little success at protecting tropical forest species from the threats they face.[33–35]

Its tropical forest focus and considerable financial clout have rapidly propelled REDD+ into the role of potential remedy to the current biodiversity crisis.[36–41] The solution seems simple. Because of their vast and vulnerable carbon stocks that could be protected at relatively low cost (though we question this assumption later in the paper), tropical forest regions are poised to receive unprecedented funds to incentivize their protection. As these very areas are also the planet's unquestioned treasure troves of biodiversity, it seems likely that a major collateral benefit of REDD+ will be widespread biodiversity protection. But recent research has revealed that it might not be so straightforward.

First, REDD+ is likely to focus on protecting forests in regions that are most cost-effective for reducing carbon emissions.[37,42] An important determinant of the biodiversity benefits of REDD+ will therefore be the degree to which biodiversity and carbon priorities overlap. At the global scale, carbon and biodiversity are correlated.[39,43] However, if deforestation rates and the opportunity costs of implementing REDD+ are considered when identifying global priorities for carbon, this correlation breaks down, and REDD+ appears likely to perform poorly for biodiversity.[37] Perversely, this lack of coincidence could even lead to biodiversity-negative results, if REDD+ displaces deforestation effort from its priorities into high-biodiversity locations.[42] It is worth noting, however, that early REDD+ activities do show a preference for high-biodiversity locations,[2,6] though this may be due to the fact that early project developers are often organizations working primarily for biodiversity conservation.[2] Additionally, a regional-scale analysis found greater congruence between REDD and biodiversity priorities,[36] but more work at this scale is required to determine if this is a general relationship.

Biodiversity may also miss out if REDD+ is implemented in such a way that only protects a forest's trees, which represent the bulk of its biomass carbon, but not its other species. The persistence of forest-dwelling species can be jeopardized by a range of threats other than the loss of the forest.[44–46] In the tropics, these species are often threatened by climate change,[47] altered fire regimes,[29,48] and hunting.[46,49] Thus, even if REDD+ were to act in the right places, if it does not address the other threats to forest biodiversity, it could deliver minimal outcomes for conservation.

A number of studies have also warned of the danger that REDD+ could promote the expansion of monoculture plantations of nonnative species or "carbon farms," potentially at the expense of native forests.[50–53] This would undoubtedly circumvent any potential biodiversity benefits and, in many cases, probably lead to biodiversity declines.[54,55] A final issue is the risk that REDD+ operates on a shorter timeframe than necessary for protecting biodiversity in the long-term, especially long-lived species.[56]

To avoid these deleterious biodiversity outcomes, conservation scientists have called for the UNFCCC to adopt strict biodiversity safeguards.[40] And this is exactly what happened at the UNFCCC's Conference of the Parties in Cancun, 2010. While the exact mechanism for implementing the decision remains undecided, Annex I paragraph 2 of the Cancun decision (Dec. 1/CP16) states that REDD+ must incentivize "the conservation of natural forests and biological diversity" and "not be used for the conversion of natural forest areas." Moreover, it must be "consistent with [. . .] relevant international conventions," which includes the Convention on Biological Diversity. In our opinion, these decisions are an important step toward putting to rest the issue of biodiversity negative outcomes from REDD+.

Promoting biodiversity-positive outcomes, such as targeting biodiversity hotspots for protection, would incur little extra cost within an international REDD+ scheme.[37] Two main options exist for capitalizing on this opportunity. The first is to develop biodiversity credits by monitoring, reporting, and verifying biodiversity outcomes from REDD+ actions[57] and linking these credits to biodiversity offsetting markets or funds.[58,59] The second is to harness biodiversity conservation funds, which exceed \$1.5 billion annually,[60] to help finance REDD+ implementation in biodiversity priorities. Whatever the mechanism, it seems that achieving the best outcomes for biodiversity from REDD+ will take some extra effort and financing to realize.

What's different about REDD+?

The importance of tropical forests as carbon sinks and biodiversity hotspots has long been recognized,[30,31,61] and the fight to save them was not born with REDD+. Reforestation and afforestation in the tropics have been a sanctioned activity under Kyoto's CDM since 2001, yet a mere 0.54% of CDM carbon credits come from forests.[62] Moreover, while strategies employed to protect biodiversity in the tropics can be effective,[63] they have failed to arrest widespread declines in forest area[17,64] and metrics of biodiversity.[65] Do we have reason to think that REDD+ will be any different?

One of the most compelling reasons for renewed hope is the level at which REDD+ is currently, and likely to continue, generating financing for tropical forest conservation. An early projection estimated that REDD+ could generate $10–60 billion per year in payments to developing countries if deforestation were to be halved.[4] At the time, the prediction seemed outlandish. But today, billions of dollars are already pouring into REDD+ readiness and early-action initiatives. More significantly, Annex I countries agreed, in the Copenhagen accord, to mobilize $30 billion in the period 2010–2012 and $100 billion per year by 2020 for climate mitigation and adaptation in developing countries, much of which could to be used for REDD+.[66] In contrast, data from the early 1990s indicate that the total protected area budget for developing countries was $0.7 billion.[67] Clearly, regardless of the exact figures, REDD+ represents a major financial boost for conserving tropical forests.

But there's more to it than simply bankrolling business-as-usual conservation. When REDD+ reaches full implementation, it is expected that payments for carbon credits will only be made when participants have demonstrated, through an approved MRV process, that emissions have been successfully reduced. Payments being conditional on performance are typical of carbon offsetting schemes, but this represents largely new ground for forest conservation. For instance, since 1989, $5.8 billion has been funneled through Indonesia's reforestation fund to promote forest recovery on degraded lands and forests. However, most of these funds have been lost to fraud or wasted on ineffectual projects, partly because payments were made without holding recipients accountable for outcomes.[68] Another difference is the time horizon of REDD+ projects. Most early REDD+ projects have expected time horizons of 20–30 years,[2] whereas the average World Bank–funded biodiversity project lasts only 6.5 years.[69] Moreover, the Copenhagen accord recognizes the need for financing to be "adequate, predictable and sustained."[66] The conditionality of payments and the long-term nature of projects give reason to believe that the financial incentives provided by REDD+ may catalyze greater change per dollar invested than past initiatives.

Challenges and limitations

Technical and implementation issues

REDD+ is still in its learning-while-doing phase, and many of the design and implementation methods are still being developed and trialed. This is not compromising financing at present, as most initiatives are currently funded through voluntary contributions, without rigorous demands on maximizing and accurately quantifying realized emissions reductions. But when the time comes for REDD+ credits to be traded on international carbon markets, the ability to convince markets that emissions reductions are real and lasting, which largely hinges on technical methods of implementing actions and monitoring outcomes, will determine market demand, and in turn, the price of REDD+ credits.[70]

Determining additionality. One of the founding principles of carbon offsetting is that carbon credits are only awarded for activities that are additional, meaning they reduce emissions to a level lower than would have occurred without the activity. In REDD+, the counterfactual scenario of "what would occur" to forests without intervention is known as the reference level, though this level could also include negotiated adjustments to fine-tune how carbon credits are generated.[71] Reference levels act as the yardstick for measuring REDD+ outcomes and can be set at subnational scales, as is often the required by early REDD+ projects, or national scales, which is the anticipated scale for REDD+ accounting.[3] Because they are the basis for generating carbon credits, reference levels determine of the potential distribution of REDD+ payments and participation as well as the perceived integrity and equity of carbon credits. This makes defining these levels one of the most important and contested aspect of any REDD+ program.

The problem is that reference levels are determined *ex ante*, and predicting the future can be a perilous undertaking. At present, there is no set method for developing reference levels; instead, a range of options have been proposed.[72–74] Most proposals involve extrapolating forward recent past trends in forest loss. At the national scale at least, recent historic emissions do a reasonable job of predicting near-term future emissions.[72] But basing reference levels solely on historic emissions has three major drawbacks. First, national or regional circumstances can change, causing unanticipated deviations from historic trends. For instance, according to national forest statistics,[18] Indonesia doubled its forest carbon emissions in the period 2000–2010 compared to the 1990s. If REDD+ had been implemented in 2000 using a 10-year historic reference level, Indonesia would have been short-changed \$18 billion over a 10-year period, assuming it could reduce its forest carbon emissions and receive compensation at a CO_{2e} price of \$10. The second drawback is that a historic emissions approach rewards countries pursuing rapid deforestation and punishes countries that have maintained high levels of forest cover and low levels of deforestation, which is ethically questionable. Third, offering high-forest, low-deforestation countries little incentive to participate in REDD+ risks accelerating deforestation in these areas as internationally mobile industries switch their investments into areas unrestricted by REDD+ policy measures.

To circumvent these drawbacks, it has been proposed that reference levels be set based on forest carbon stocks, not fluxes.[75] Under such a scheme, countries that have lots of forest but little forest loss, like Suriname and Gabon, could receive carbon credits for their forests that would otherwise be overlooked under a history-based scheme.[72] While it may be ethically sound to reward countries that have exercised good forest stewardship, it is questionable whether paying to protect forests that are not under threat, at least in the near-term, has any additional climate benefits.[76]

Clearly, how reference levels are set can either lead to short-changing participants, as with our Indonesian example, or wasted funds. Participants whose reference levels are set too high will be able to generate credits without taking additional action to reduce emissions. Because these emissions reductions will incur no or low implementation and opportunity costs, they will be the most price competitive on carbon markets. Such falsely generated credits could dominate carbon markets, especially if markets are fairly small.

At present, the best way forward for setting reference levels appears to be to combine incentives to reduce high levels of deforestation with incentives to maintain low rates of forest loss.[77,78] Modeling shows that a combined approach would maximize participation and minimize the costs of emission reductions by reducing the potential for increased deforestation in nonparticipating countries.[73] Additionally, historic trends could be further modified by modeling land-cover change to anticipate deviations from this trend, though the predictive accuracy of such models leaves much to be desired.[79] Finally, reference levels are unlikely to be determined by data alone, but also through political negotiation between participating countries trying to carve out a decent slice of REDD+ revenues and Annex I countries trying to maximize the integrity of emission reductions. The impact of these negotiations on scientifically justified reference levels should be minimized.

Permanence and leakage. Since its inception, REDD+ proponents have been challenged to convince investors that interventions will lead to lasting climate benefits.[80,81] Skeptics fear that forest carbon protected by REDD+ activities could simply be released into the atmosphere at a later date through land clearing, climatic change, or fire, undermining the intended climate benefits.[48,82] The perceived lack of permanence—the longevity of a carbon pool and the stability of its stocks—in emissions reductions could seriously undermine demand for REDD+ credits.

Permanence has been a particularly important issue for early REDD+ projects, as they typically protect a particular tract of forest,[2] which can be vulnerable to encroachment and stochastic events.[33] The transition of REDD+ from site-based projects to national-scale implementation changed this considerably. When operating at the national scale, the MRV process is based on the realized rate of net national forest-carbon emissions, not whether a particular tract of forest is protected. This makes reducing emissions from forests appear identical to reducing the rate of fossil fuel emissions, where rates of emissions are reduced, not particular fossil fuel reserves. It is therefore argued that, like efforts targeting fossil fuel emissions, REDD+ will lead to permanent benefits, as even if countries opt out in the future, emissions should simply return to the pre-REDD+ rates and not undue previous reductions.[4,83]

However, comparing historic trends in deforestation with those of fossil fuel emissions raises doubts about the similarity of the two sources. For instance, aside from a slight increase each year, fossil fuel

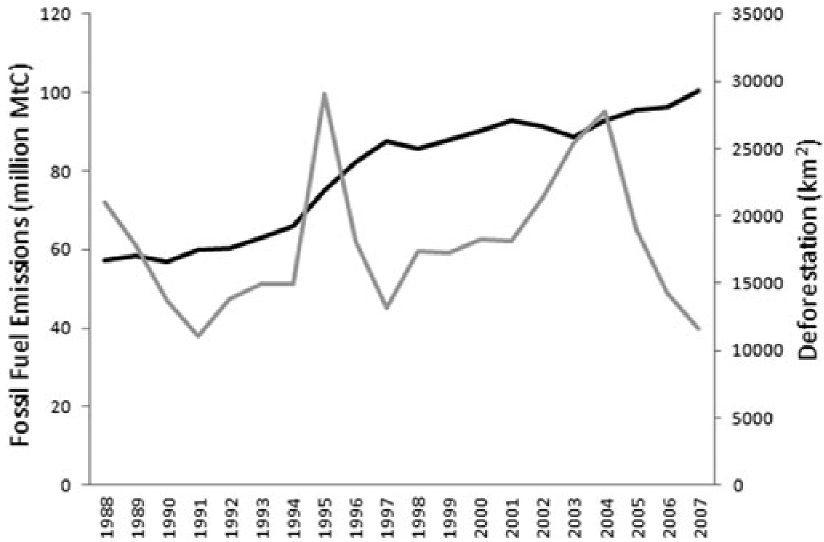

Figure 1. Annual fossil fuel emissions (black line) and deforested area (gray line) for Brazil, 1988–2007. Data are from the Brazilian National Institute of Space Research (http://www.inpe.br/ingles/) and the Carbon Dioxide Information Analysis Centre (http://cdiac.ornl.gov/ftp/trends/emissions/bra.dat).

emissions in Brazil show little interannual variation (Fig. 1). On the other hand, deforestation is marked by a series of peaks and troughs, determined largely by national and international economic conditions and government policies and investments.[84] Given the capacity for deforestation rates to increase over short periods of time, there is a risk that REDD+ achievements are undone by accelerated deforestation following periods of low forest loss.

A number of options exist to guard against the risk of nonpermanence. These include withholding credits from the market to act as a buffer or insurance policy against future emissions, discounting the value of credits, and creating temporary instead of permanent credits. These options have the effect of either paying less per ton of averted emissions or delaying payments. Both options will reduce the cost-competitiveness of REDD+ compared with the drivers of forest loss, and how to strike the right balance between reducing the risks of nonpermanence and sufficiently rewarding participants for emissions reductions remains to be determined.

REDD+ carbon credits could also be undermined by the effects of leakage, which is the shifting or displacement of deforestation from an area under REDD+ restrictions to an area without these restrictions.[85] The clearest empirical evidence for leakage comes from work in the Peruvian Amazon. By observing changes in forest loss rates inside and outside of protected areas both before and after park establishment, Oliveira *et al.*[86] were able to show that while deforestation rates decreased inside parks, these benefits were undermined by increased deforestation in the adjacent areas. Even if leakage is unavoidable, the integrity of REDD+ credits could be maintained if leakage is accurately quantified. However, to quote from the leakage assessment for the Noel Kempff climate action project, "through the process of assessing leakage [. . .] it has become ever more apparent that quantifying leakage is not a straightforward process."[87]

As with the issue of permanence, the transition from site-based projects to national-scale implementation has considerably changed the potential impacts of leakage on REDD+ activities. The national-scale MRVing of carbon credits implicitly accounts for any within-country leakage of deforestation effort, effectively side-stepping the need to do so explicitly. Still, the drivers of deforestation are increasingly able to shift from one country to another,[88] raising the possibility of leakage occurring across national borders. In fact, economic modeling suggests that when logging is reduced in one country, 42–95% of this reduction may be leaked as increased logging in other countries.[89] International leakage could be minimized by encouraging all developing countries to participate in REDD+,[90] partially through offering attractive reference levels to high-forest, low-deforestation countries.

Monitoring forest carbon. To set reference levels and track progress, REDD+ programs must monitor and report on changes in forest cover and carbon stocks through time. With the widespread availability of remotely sensed data—LANDSAT has been producing data continually since 1972—and ongoing national forest census programs,[91] it would seem as though monitoring forests would not present a major hurdle for REDD+ implementation. However, in 2009, only 3% of countries had the capacity to monitor and report on changes in forest cover for REDD+,[92] and still no global forest monitoring network exists.[93] Even more concerning are the findings by Pelletier *et al.*[94] that, in Panama, deforestation would have to be reduced by half before it could be detected above the levels of statistical uncertainty from using available data sources. Most of the uncertainty in quantifying forest carbon stocks surrounds our ability to map the spatial variation of carbon within a given land-cover type, such as primary forests.[94] The problem is that most optical remote sensing technologies are unable to determine carbon variation above a fairly low carbon value, and ground-based techniques are too slow and expensive to employ across vast areas.[91]

The limitations of current forest monitoring programs has not gone unnoticed, and in 2010–2011 $6.3 billion was spent on monitoring land carbon,[95] with some areas of progress showing considerable potential. Notable among them are recent advancements in using airplane-based LiDAR (light detection and ranging) technology in conjunction with field surveys and satellite-based optical remote sensing. This hybrid approach has been used to map the carbon content of 4.3 million hectares of the Amazon basin with high accuracy and precision and at a cost of only $0.08 per hectare.[96] While it may be cost-effective, using LiDAR combined with other remote sensing data is technically demanding, and whether this approach can be scaled up and replicated to deliver wall-to-wall mapping remains to be seen.

Socioeconomic and political issues

A REDD+ mechanism would require decision makers to commit to long-term decisions that might have implications that extend far beyond the immediate REDD+ affected areas and communities. Such impacts might include not only lost financial opportunities from withholding conventional land use activities, but also less-direct and tangible socioeconomic costs associated with REDD+. Therefore, the success of REDD+ implementation requires careful consideration of the full range of economic, social, and political implications of avoiding deforestation.[97]

The most obvious examples of such costs include the loss of employment and revenue generation from raw material processing and other value-adding downstream industries. The Malaysian oil palm industry, which continues to threaten the country's remaining rainforests, particularly in Borneo, contributes about 5–6% of Malaysian GDP and provides direct employment for 570,000 workers. The industry is associated with a further 830,000 workers employed in downstream activities. Oil palm exports also generate foreign exchange earnings amounting to around $10 billion annually. These funds contribute to rural development including the improved provision of piped water, electricity, communications, roads, schools, and healthcare through land schemes coordinated by the Federal Land Development Authority.[98] Any downscaling of oil palm activities and associated industries could lead to significant socioeconomic impacts, manifest in the form of constrained opportunities for regional economic growth, including reduced tax revenues for regional and local governments. Governments might become less inclined to invest in such regions, particularly if REDD+ forfeits any opportunity for development over a period of 20–30 years. Consequently, compared to a non-REDD+ area, REDD+ regions might have poorer infrastructure and telecommunications, limited mobility and market access, and lower-quality public services. Recognizing this issue, some forest-rich nations are already advocating for REDD+ funds to be allocated to development outside the forestry sector. Notably, Guyana specifically calls for REDD investments in private sector entrepreneurship, electricity, roads, health, and education.[99]

REDD+ might also lead to less tangible but equally important political and socioeconomic costs relating to national and local development. The future livelihood options of local communities might be constrained, as is arguably the case with protected area systems.[100] Perhaps more importantly, forest-using communities with unclear land tenure might be denied access to the lands they manage and depend on, leading to conservation refugees.[101] At the national level, REDD+ implementation might encourage the migration of people from REDD+ affected areas to urban centers. Such demographic changes could raise political and social concerns and further undermine government priorities for investment in depopulated rural areas. By promoting national institutions charged with forest management, there is also the risk that REDD+ will reverse decades of work toward decentralized forest governance.[102,103]

Perhaps the least-considered aspect of REDD+ is its potential implications for international relations. A REDD+ mechanism that limits the supply of key commodities could alter conventional political ties where such relations are strongly based on trade. For example, Japan's historical relationships with Indonesia and the Philippines have largely been based on the trade of natural resources such as timber.[104] A central priority of Japan's foreign development aid has been to secure the development and import into Japan of natural resources.[97] Such consumer–client relationships, as well as other political and business ties that shape political and economic interactions between countries, might limit the widespread implementation of REDD+.

While there are important risks that REDD+ could constrain development options and lead to widespread and possibly unanticipated socioeconomic costs, as detailed above, these risks have been at least partially mitigated by the addition of socioeconomic safeguards into REDD+ negotiations (Cancun Decision 1/CP.16). These safeguards include the requirement that activities are "undertaken in accordance with national development priorities" and "in the context of sustainable development and reducing poverty." What remains is identifying how REDD+ can best be implemented to fulfill both development and climate mitigation objectives.

Norway–Indonesia REDD+ agreement

In May 2010, the governments of Norway and Indonesia signed a Letter of Intent for a REDD+ partnership that would contribute to significant reductions in greenhouse gas emissions from deforestation, forest degradation, and peatland conversion in Indonesia.[105] Phase 2 of this agreement requires Indonesia to implement a two-year moratorium to suspend all new concessions for the conversion of peat and natural forests. The wider goal of this moratorium is to create a baseline on critical elements of forests, peatlands, and "degraded lands" that is strategic to the effective implementation of a nationwide REDD+ strategy in the future. This partnership offers a good opportunity for Indonesia to mitigate its deforestation and carbon emission levels. However, it has also galvanized intense debates within Indonesian society regarding the scope of the moratorium and its consequent environmental and socioeconomic ramifications.

After much political wrangling between the Indonesian Ministry of Forestry and the country's REDD+ Task Force, the moratorium was signed into effect in May 2011 by Indonesian President Dr. H. Susilo Bambang Yudhoyono (Presidential Instruction No. 10/2011). An accompanying first draft of an "indicative map" displays land areas protected by the moratorium. Critics point out that the lack of clarity and transparency about how the indicative map was generated makes it difficult to assess its efficacy for reducing greenhouse gas emissions.[106] Environmentalists are also disappointed with the fact that large swaths of secondary forests, as well as any forest that has already been offered up to plantation companies as concessions, will be exempted from the moratorium.[107] On the other hand, some community and village forestry development projects are now under threat of being abandoned because such activities do fall within the remit of the moratorium. Social nongovernmental organizations argue that the moratorium is biased toward big agribusinesses while severely threatening ongoing efforts at developing sustainable forestry programs at the grassroots level.[108]

Putting aside the controversies about the moratorium, the success of the Norway–Indonesia partnership, and Indonesia's emissions reduction efforts in general, will depend on its ability to overcome its patchy record in forest conservation.[109] A ban on the expansion of oil palm plantations on peatlands in 2007 was repealed only two years later to allow the conversion of 2 million ha of peatlands.[110,111] Furthermore, despite promises by the Indonesian government to reduce forest fires, the number of human-induced fires continues to increase.[112] Most crucially, the Indonesian government has been criticized for rampant and systematic corruption. The country's Reforestation Fund has allegedly been used for politically expedient projects unrelated to forest restoration.[68]

The Norway–Indonesia REDD+ agreement faces tremendous challenges. But the authorities can ill-afford to allow this initiative to fail. Many observers regard the Norway–Indonesia partnership as a harbinger of outcomes in other forest-rich nations seeking REDD+ funds, such as Papua New Guinea, Democratic Republic of Congo, and Colombia. Therefore, the Indonesian case study will likely have profound implications for climate change mitigation and conservation across the developing tropics.

Way forward

Clarifying and examining fundamental assumptions of REDD+

One of the fundamental assumptions of REDD+ is that the lack of financial incentives to protect forests is the primary reason behind deforestation and forest degradation in forest-rich developing nations. However, tropical deforestation is, in fact, often driven by a complex suite of social, political, and economic factors, including corruption, weak governance, human migration, and national and international political pressures. As we discussed above, even if REDD+ is able to fully compensate for the direct financial opportunity costs of avoided deforestation, there still might be other indirect and less tangible factors that would continue to drive forest conversion and undermine REDD+. Therefore, the expectation for a well-funded REDD+ to fully deliver on avoiding deforestation might have to be tempered with the realization it must either be part of a portfolio of solutions for mitigating forest and carbon loss or further expanded to include such solutions.

In the past few decades, without the help of REDD+, a number of countries transitioned from net deforesters to forest-gaining countries, including China, India, Vietnam, the Republic of Korea, and Chile. What drove these transitions can provide lessons for how best to compliment REDD+ financial incentives with additional socioeconomic and political reforms. Most notably, the transition from forest-losing to forest gaining was characterized by rapid economic development, an increase in the perceived value of wood, a strengthening of the rights of indigenous and forest-using peoples, and the establishment of aggressive afforestation and reforestation programs.[113] In addition to putting a price on forest carbon, incentives to stimulate these changes could also include stimulating nonforestry sectors of the economy, increasing the value of wood by increasing in-country downstream processing, and tackling ongoing indigenous land rights disputes.

There has been a tendency in the scientific literature and by political and environmental lobbyists to view REDD+ not strictly as a tool for mitigating climate change, but rather as a silver bullet for tropical forest conservation. Clearly, the opportunities to derive cobenefits resulting from avoided deforestation appear too important to pass up. Nevertheless, there is a real risk of overburdening REDD+ with too many good intentions, and emission reductions must therefore remain the main purpose of REDD+. The challenge ahead is to develop complimentary mechanisms and markets to allow for the bundling of benefits, such as biodiversity conservation, without increasing transaction costs of participating in REDD+ carbon markets.

It is also important to consider the dilemma posed by REDD+ for climate change mitigation. At the UN climate negotiations in Copenhagen in 2010, developed countries, including European Union members, pledged to reduce emissions by up to 20% by 2020. The more successful that REDD+ might eventually become at supplying carbon credits to Annex I countries to meet their emissions targets, the less incentives there would be for domestic actions to cut emissions.[95] Therefore, somewhat paradoxically, to achieve effective emissions reductions globally, either ceilings will have to be imposed on the use of REDD+ credits for offsetting emissions in developed countries, or separate REDD+ targets will need to be set.

Conclusion

In our opinion, REDD+ is currently the most promising mechanism for conserving tropical forests, largely because of its unprecedented ability to harness funds and put them to use in ways that are fundamentally more effective than past efforts. But to the dismay of some, after six years of negotiations, an officially endorsed REDD+ mechanism is yet to be finalized. At the same time, this delay presents an opportunity to further shape REDD+ to address persistent issues and create new opportunities. To ensure that REDD+ acts as a true game changer in the struggle to protect tropical forests, and not just a fleeting quick-fix, improvements should focus on REDD+'s ability to provide or compliment a suite of nonmonetary incentives to encourage developing nations to transition away from net deforestation while also accessing low transaction cost and high-volume carbon markets or funds, and align with, not undermine, a globally cohesive attempt to mitigate climate change.

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Conflicts of interest

The authors declare no conflicts of interest.

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